

GANESH GUPTA

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Experience

Oracle

Jun 2025 – Present

Software Engineer, AI Platform

- Built and maintained Airflow DAGs orchestrating multi-stage data ingestion and transformation pipelines across AWS and GCP, enabling scheduled ETL workflows with dependency management and automated failure alerting.
- Designed BigQuery dimensional schemas and optimized analytical queries for large-scale data marts, reducing compute costs through partition pruning and clustering while cutting p99 query latency 420ms to 140ms.
- Integrated generative AI and LLM interfaces (OpenAI Agents SDK, MCP tool-calling) directly into data workflows, automating exploratory analysis and surfacing efficiency signals across millions of daily transactions.

QureAI

Jan 2024 – May 2025

ML Infrastructure Engineer

- Owned the data pipeline feeding production model evaluation end-to-end: sourcing, transformation, and quality control on a 2K-query eval set, lifting retrieval recall 61% to 89% via improved data curation.
- Built PySpark data processing jobs for large-scale feature and dataset preparation, and a production observability framework tracking data quality, latency, and regressions across pipeline versions.

Nomura Research Institute Financial Technologies

Aug 2020 – Aug 2023

Software Engineer

- Designed high-throughput SQL data pipelines for financial data ingestion and aggregation, cutting batch processing from 8 minutes to 90 seconds via indexed views, partitioning, and lock contention elimination.
- Architected and operated 25+ production Java microservices sourcing, transforming, and serving data via REST APIs for 1M+ users, sustaining 99.9% uptime over 3 years on AWS (EC2, RDS, SQS) with relational and NoSQL data stores.
- Delivered metrics dashboards and technical documentation translating pipeline health into actionable insights for cross-functional stakeholders, reducing production escalations 60% in one quarter.

Projects

Distributed KV Store / VPC Control Plane Simulator | github.com/ganeshhgupta/distributed-kv-store

- Built a Raft-consensus key-value store in-house with leader election, log replication, and log compaction; companion VPC control plane simulator modeling subnet routing and security group enforcement for fault-tolerant distributed coordination.

Watch-your-LLM | github.com/ganeshhgupta/watch-your-llm

- Production performance and eval tooling with rubric-based scoring, drift detection, and per-call latency benchmarking across model versions; FastAPI backend, PostgreSQL.

Education

University of Texas at Arlington

Aug 2023 – May 2025

MS, Computer Science, GPA 4.0/4.0

MCKV Institute of Engineering

Jul 2016 – Jul 2020

BTech, Computer Science

Skills

Languages: Python, SQL, Java, TypeScript

Data Engineering: ETL/ELT Pipelines, Airflow, PySpark, BigQuery (Data Warehousing), Data Modeling, Data Quality & Validation

Real-Time & APIs: Kafka, SQS (Distributed Messaging), FastAPI, REST APIs

Cloud & Infra: AWS (EC2, Lambda, S3), Azure, GCP (BigQuery, Cloud Run), Docker, Kubernetes, CI/CD, Git

Certifications

AWS Certified Developer Associate | GCP Professional MLOps Engineer